Quantification: I534_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	34
interior lacuna count	23
frame-edge lacuna count	11

Interior lacunae only: mean, median and sample SD (n = 23)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2191.48	1992	1746.99	px^2
major_axis_length_px	lacuna shape	71.9492	64.4016	34.0001	px
minor_axis_length_px	lacuna shape	38.8075	35.6451	16.1409	px
aspect_ratio	lacuna shape	1.9603	1.8104	0.8755	unitless
eccentricity	lacuna shape	0.7773	0.8336	0.167	unitless
solidity	lacuna shape	0.8049	0.8022	0.1004	unitless
roots_count	headline	6.8261	5	4.3657	count
ring_length_r30_px	headline	223.696	193	111.493	px
ring_length_r60_px	reported	460.13	431	198.962	px
owned_length_px	ownership-dependent	565.562	347.309	646.767	px
edge_count	ownership-dependent	27.4348	17	31.383	count
mean_edge_length_px	ownership-dependent	21.0822	20.782	5.7257	px
perimeter_px	normalised	230.807	215.037	107.749	px
ring_area_r30_px2	normalised	8152.26	7039	3079.73	px^2
ring_area_r60_px2	normalised	17290.2	18445	6428.62	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9935	1	0.0204	unitless
ring_density_r30	normalised	0.0282004	0.0272545	0.0109325	px^-1
ring_density_r60	normalised	0.0275031	0.0250495	0.00858745	px^-1
roots_per_100px_perimeter	normalised	2.9418	2.899	1.2563	per 100 px
ring_attached_length_r30_px	network v2	190.957	179	112.78	px
ring_attached_length_r60_px	network v2	319.652	284	199.086	px
ring_length_w_r30_px	network v2	254.405	215.924	128.167	px
ring_length_w_r60_px	network v2	523.577	486.233	226.226	px
sholl_crossings_r10	network v2	5.3043	5	2.6532	count
sholl_crossings_r20	network v2	6.1739	5	3.5758	count
sholl_crossings_r30	network v2	6.7826	6	3.2885	count
loop_count	loops, report only	1.5217	0	3.1023	count
loop_edge_share	loops, report only	0.1048	0	0.142	share
ring_width_mean_r30_px	width	7.5833	7.7291	0.8581	px

Per field
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measure	value	units
analysed_area_px2	912415	px^2
total_skeleton_length_px	23451	px
canalicular_length_density_per_px	0.0257021	px^-1
junction_count	471	count
junction_density_per_px2	0.00051621	px^-2
lacuna_count	34	count

lacunae_per_px2	3.726e-05	px^-2
skeleton_component_count	302	count
mean_component_length_px	77.6523	px
median_component_length_px	27.5	px
field_length_density_w_per_px	0.0292138	px^-1
field_density_without_flagged_per_px	0.0267597	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	297	px
unattached_component_count	20	count
canalicular_length_density_attached_per_px	0.0257021	px^-1
field_length_density_with_unattached_per_px	0.0260276	px^-1
loop_count	44	count
loop_edge_share	0.1257	share
edge_count	1241	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0291	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.3137	px
width_pixels_used	22394	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6611	188.219	59.1114	3.1841	0.9494	0.7758	-1.5406	20.9109	224.092	True
2	1077	56.963	28.6202	1.9903	0.8646	0.7016	0.9022	20.0604	411.138	True
3	10729	252.088	59.2099	4.2575	0.972	0.8231	-1.467	29.4163	690.039	True
4	7272	192.335	53.4975	3.5952	0.9605	0.8264	1.5113	27.3416	895.513	True
5	485	30.2188	22.2379	1.3589	0.6771	0.8692	-1.007	80.699	192.295	False
6	1175	57.981	35.1354	1.6502	0.7955	0.6155	-1.5521	82.0417	147.133	False
7	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
8	16316	190.924	121.711	1.5687	0.7705	0.7618	-1.2586	206.568	82.4135	True
9	571	49.2556	22.5437	2.1849	0.8891	0.6798	-0.4492	178.853	461.693	False
10	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
11	506	33.0898	22.5781	1.4656	0.731	0.8447	-0.2309	223.868	412.781	False
12	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
13	441	59.6262	11.5994	5.1404	0.9809	0.7513	-0.9334	296.8	894.338	False
14	5138	135.156	50.3775	2.6829	0.9279	0.8874	-0.8946	375.376	523.812	False
15	749	51.0208	20.659	2.4697	0.9144	0.8874	1.1423	366.493	18.0734	True
16	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
17	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
18	623	33.6165	26.9149	1.249	0.5991	0.8534	-1.4105	448.144	1010.89	True
19	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
20	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
21	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
22	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
23	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
24	669	36.2679	35.241	1.0291	0.2363	0.673	1.4264	577.549	741.005	False

25	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
26	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False
27	812	45.0748	28.6241	1.5747	0.7725	0.7785	0.6665	738.431	13.2463	True
28	551	40.1987	20.8605	1.927	0.8548	0.7951	1.0759	753.867	437.468	False
29	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
30	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False
31	38322	496.242	167.137	2.9691	0.9416	0.5887	-1.5553	972.902	712.084	True
32	526	35.2616	23.4896	1.5012	0.7458	0.7525	1.0779	860.238	1011.11	True
33	2059	64.4016	53.6924	1.1995	0.5522	0.708	-1.4007	899.013	923.481	False
34	569	32.4583	24.6996	1.3141	0.6488	0.8493	-0.8213	949.207	269.23	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	249	364	350.321	23	15.2313	7.1564
2	10	250	375	738.642	45	16.4143	7.3007
3	8	259	581	157.196	10	15.7196	7.2057
4	3	184	418	134.125	6	22.3541	8.1476
5	3	96	183	86.9411	8	10.8676	8.1198
6	6	136	139	134.74	10	13.474	5.5822
7	2	193	368	68.4264	4	17.1066	8.307
8	26	774	1212	1609.34	75	21.4578	7.2443
9	5	271	503	2951.04	142	20.782	7.9594
10	2	52	93	60.7696	5	12.1539	6.2562
11	5	274	539	322.764	14	23.0546	7.7309
12	6	265	645	261.309	13	20.1007	7.7291
13	5	166	292	234.309	9	26.0343	6.9689
14	13	318	660	1008.68	49	20.5853	7.8377
15	3	126	262	183.61	6	30.6017	6.8891
16	13	460	997	1583.83	76	20.8399	7.061
17	11	334	463	330.593	21	15.7425	7.3598
18	4	130	342	352.049	14	25.1463	7.5961
19	5	172	447	347.309	19	18.2794	8.499
20	4	151	431	180.095	10	18.0095	8.0738
21	18	496	814	820.411	50	16.4082	7.4463
22	7	272	601	771.625	32	24.1133	7.6143
23	13	311	542	467.747	21	22.2737	7.8907
24	11	204	369	386.25	23	16.7935	6.8089
25	3	147	410	421.889	11	38.3536	8.8798
26	4	93	231	169.61	8	21.2013	7.4866
27	4	68	135	49.5563	5	9.9113	5.5824
28	4	190	537	361.25	17	21.25	7.9135
29	6	190	379	585.73	24	24.4054	7.2164
30	3	108	403	278.321	9	30.9245	7.5782
31	19	588	1086	696.571	36	19.3492	8.1042
32	3	140	263	45.2843	4	11.3211	6.9673
33	9	248	385	1186.7	59	20.1136	8.9617
34	1	50	245	48.3553	2	24.1777	5.3905

Normalised per lacuna

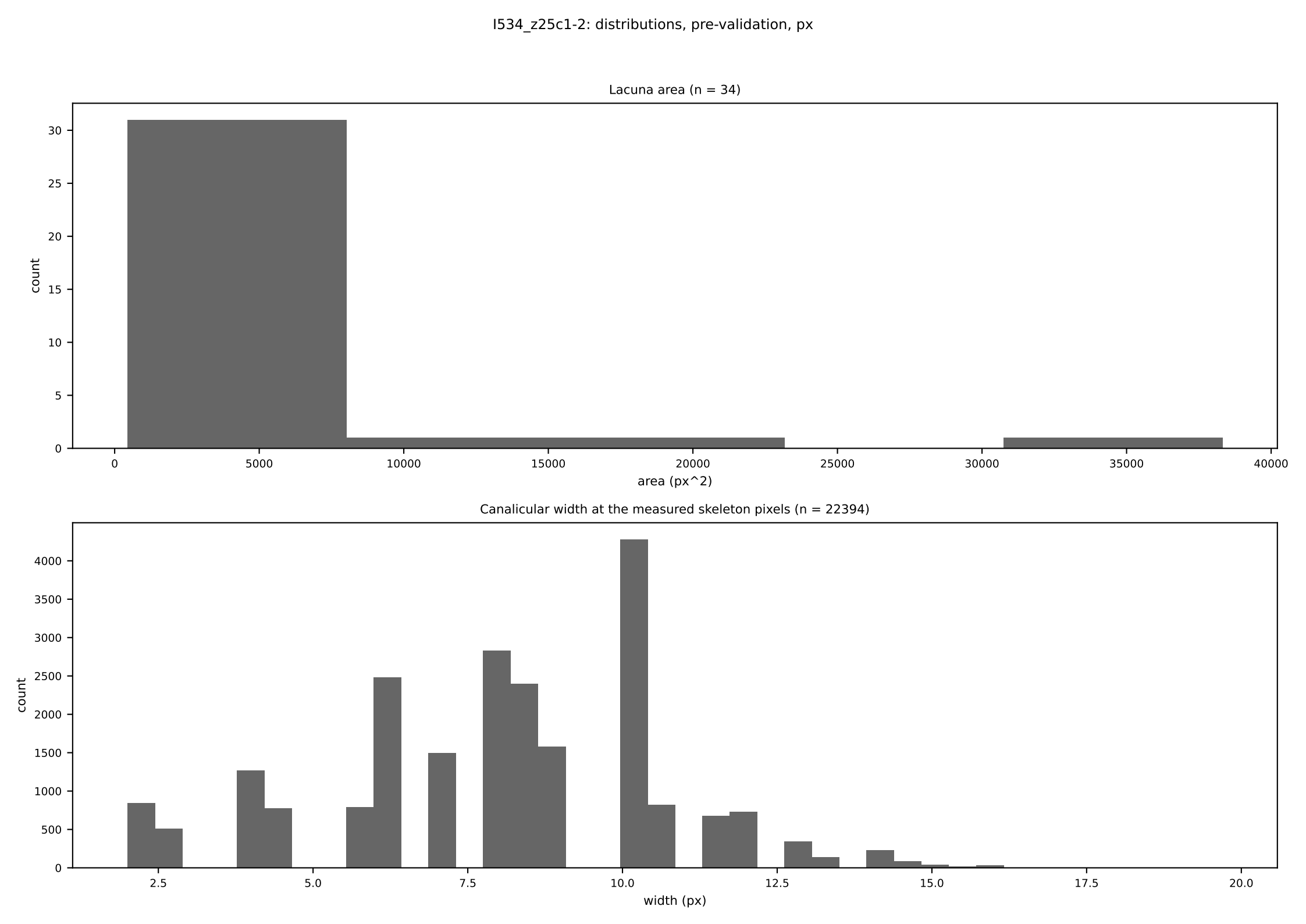
lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	685.506	7928	12802	0.596	0.572	0.0314077	0.0284331	1.167
2	192.167	4809	7186	0.6966	0.6322	0.0519859	0.0521848	5.2038
3	751.654	12022	23386	0.6838	0.6378	0.0215438	0.0248439	1.0643
4	629.754	9543	16652	0.6904	0.571	0.0192812	0.0251021	0.4764
5	99.1543	2463	4527	1	1	0.0389769	0.0404241	3.0256

6	206.965	4990	6604	1	0.9867	0.0272545	0.0210478	2.899
7	132.125	5891	11356	1	1	0.0327618	0.0324058	1.5137
8	930.744	18415	31353	0.8227	0.7587	0.0420309	0.0386566	2.7935
9	133.054	5762	11142	1	1	0.0470323	0.0451445	3.7579
10	237.965	7078	14276	0.7631	0.6986	0.00734671	0.00651443	0.8405
11	107.154	5112	12725	1	1	0.0535994	0.0423576	4.6662
12	334.475	11891	25749	1	1	0.0222858	0.0250495	1.7939
13	128.853	6312	14219	1	1	0.0262991	0.0205359	3.8804
14	366.517	12817	22622	1	1	0.0248108	0.0291751	3.5469
15	128.947	4467	8594	0.694	0.6341	0.0282068	0.0304864	2.3265
16	419.345	11919	23456	1	1	0.0385938	0.0425051	3.1001
17	332.676	11820	20074	1	1	0.0282572	0.0230647	3.3065
18	114.184	3796	10383	0.6248	0.5844	0.0342466	0.0329385	3.5031
19	249.078	9533	20117	1	0.9067	0.0180426	0.02222	2.0074
20	262.877	9914	22390	1	1	0.015231	0.0192497	1.5216
21	405.588	13364	31075	1	1	0.0371146	0.0261947	4.438
22	278.09	9508	20761	1	1	0.0286075	0.0289485	2.5172
23	406.416	12535	23662	1	1	0.0248105	0.0229059	3.1987
24	166.752	5502	11604	1	1	0.0370774	0.0317994	6.5966
25	220.35	8451	20450	1	1	0.0173944	0.0200489	1.3615
26	153.238	5825	12202	1	0.9659	0.0159657	0.0189313	2.6103
27	188.368	3275	4996	0.6244	0.5829	0.0207634	0.0270216	2.1235
28	121.054	5947	17492	1	1	0.0319489	0.0306998	3.3043
29	215.037	8340	19166	1	0.9972	0.0227818	0.0197746	2.7902
30	146.853	6827	18445	1	1	0.0158195	0.0218487	2.0429
31	2457.27	34561	57916	0.6713	0.6268	0.0170134	0.0187513	0.7732
32	123.433	3598	6752	0.6286	0.5841	0.0389105	0.0389514	2.4305
33	316.676	7039	11620	1	1	0.0352323	0.0331325	2.842
34	106.225	5740	16217	1	0.9941	0.0087108	0.0151076	0.9414

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	243	282.794	417.747	5	5	5
2	230	355	282.38	430.262	4	6	6
3	232	387	297.007	662.156	5	7	5
4	125	176	213.995	482.416	5	5	8
5	94	126	108.012	208.924	2	2	3
6	126	127	155.125	157.953	5	4	1
7	179	327	215.924	410.635	7	5	5
8	717	1003	864.754	1364.96	16	17	21
9	240	364	313.664	576.73	8	8	8
10	52	52	63.1838	106.983	2	1	0
11	262	452	311.208	617.872	6	8	6
12	224	377	292.48	722.813	7	6	7
13	134	284	188.681	329.421	3	4	5
14	296	584	370.019	751.855	9	7	11
15	114	152	140.154	300.764	3	2	6
16	416	877	530.345	1135.38	10	16	16
17	324	453	392.161	534.487	6	9	9
18	109	303	145.397	390.191	2	3	2
19	125	276	191.196	502.961	3	7	6
20	64	125	167.054	486.233	2	4	6
21	480	694	566.801	924.879	11	15	9
22	211	392	308.179	701.21	6	9	11
23	248	365	344.522	609.73	7	6	10
24	201	254	227.267	415.12	4	2	6
25	97	176	169.196	477.416	4	6	6

26	76	125	111.225	266.764	2	4	5
27	52	52	70.0416	151.711	1	0	3
28	120	240	209.267	598.174	5	5	9
29	182	297	215.338	424.363	6	4	6
30	105	217	127.782	467.073	4	3	3
31	404	573	647.772	1194.66	14	15	13
32	103	148	162.61	303.321	4	3	4
33	177	209	278.279	437.889	4	5	4
34	11	11	57.598	284.392	1	3	4


I534_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 34)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22394)
count
width (px)